Renormalization Group Flow as Optimal Transport

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Phys. Rev. D **108**, 025003 (2023)

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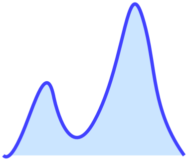
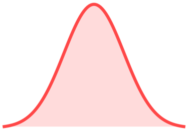
June 1, 2026

Question and Conclusion



RG coarse graining $\longrightarrow$



UV

$$P_{\mathrm{UV}} = \frac{e^{-S_{\mathrm{UV}}}}{Z}$$

IR

$$P_{\mathrm{IR}} = \frac{e^{-S_{\mathrm{IR}}}}{Z}$$

From an information-theoretic perspective, what does RG imply?

$$-\Lambda \frac{d}{d\Lambda} P_\Lambda[\phi] = -\nabla_{W_2} D_{\mathrm{KL}}(P_\Lambda[\phi] \| Q_\Lambda[\phi])$$

RG is equivalent to the optimal transport gradient flow

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Exact RG: Basic setup

Euclidean scalar field theory on $\mathbb{R}^d$, $\quad \phi : \mathbb{R}^d \to \mathbb{R}$

Probability functional

$$P_\Lambda[\phi(x)] = \frac{e^{-S_\Lambda[\phi]}}{Z_\Lambda}, \qquad \Lambda \sim 1/\ell.$$

Generating functional

$$Z_\Lambda[J] = \int [d\phi] \, \exp\left[-S_\Lambda[\phi] + \int d^d x \, J(x)\phi(x)\right].$$

Physical properties below the cutoff scale are preserved under the RG flow.

$$-\Lambda \frac{d}{d\Lambda} Z_\Lambda[J] = 0.$$

For details, refer to article or report Diffusion Model as Inverse RG.

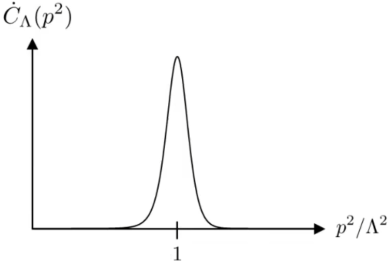
Wegner–Morris flow equations

$$-\Lambda \frac{d}{d\Lambda} P_\Lambda[\phi] = \int d^d x \, \frac{\delta}{\delta\phi(x)} \left(\Psi_\Lambda[\phi, x] P_\Lambda[\phi] \right).$$

Useful form

$$\Psi_\Lambda[\phi, x] = -\frac{1}{2} \int d^d y \, \dot{C}_\Lambda(x-y) \frac{\delta \Sigma_\Lambda[\phi]}{\delta\phi(y)},$$

$$\Sigma_\Lambda[\phi] := S_\Lambda[\phi] - 2\widehat{S}_\Lambda[\phi].$$



Wegner-Morris flow equation

$$-\Lambda \frac{d}{d\Lambda} P_\Lambda[\phi] = \frac{1}{2} \int d^d p \, \dot{C}_\Lambda(|p|) \left[\frac{\delta^2 P_\Lambda[\phi]}{\delta\phi(p)\delta\phi(-p)} + 2 \frac{\delta}{\delta\phi(p)} \left(\frac{\delta \widehat{S}_\Lambda}{\delta\phi(-p)} P_\Lambda[\phi] \right) \right].$$

Corresponding Langevin form

$$d\phi_t(p) = -\dot{C}_t(|p|) \frac{\delta \widehat{S}_t}{\delta\phi_t(-p)} \, dt + \sqrt{\dot{C}_t(|p|)} \, dW_t(p)$$

Optimal Transport (Monge formulation)

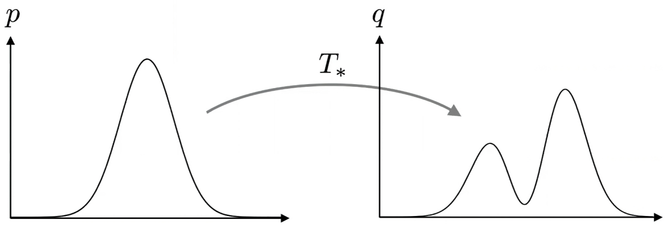


Fig. 2

Initial and final distributions: $p(x), q(x)$

Transport map: $T : X \to X,\ T_* p = q$

Cost: $c : X \times X \to \mathbb{R}$

Minimize

$$M[T] = \int_X dx\, p(x) c(x, T(x))$$

Optimal Transport (Kantorovich formulation)

Initial and final distributions: $p(x), q(x)$

Transport map: $T : X \to X,\ T_* p = q$

Cost: $c : X \times X \to \mathbb{R}$

Minimize

$$M[T] = \int_X dx\, p(x) c(x, T(x))$$

Initial and final distributions: $p(x), q(x)$

Transport kernel: $\pi(x, y)$

$$\int_X dy\, \pi(x, y) = p(x), \qquad \int_X dx\, \pi(x, y) = q(y)$$

Cost: $c : X \times X \to \mathbb{R}$

Minimize

$$K[\pi] = \int_{X \times X} dx\, dy\, \pi(x, y) c(x, y)$$

Wasserstein–2 Distance

Wasserstein–2 Distance definition

$$W_2^2(\rho_0, \rho_1) = \inf_{\pi \in \Pi(\rho_0, \rho_1)} \int_{X \times X} |x - y|^2 \, d\pi(x, y).$$

Inner product for $\eta_i = -\nabla \cdot (\rho \nabla \varphi_i)$:

$$\langle \eta_1, \eta_2 \rangle_{W_2, \rho} = \int_X \rho \, \nabla \varphi_1 \cdot \nabla \varphi_2 \, dx.$$

Gradient under this metric

$$\nabla_{W_2} \mathcal{F} = -\nabla \cdot \left(\rho \nabla \frac{\delta \mathcal{F}}{\delta \rho} \right).$$

Wasserstein–2 distance measures the minimum transport energy required to move one probability to another

Heat flow as a Wasserstein–2 gradient flow

Heat equation

$$\frac{\partial}{\partial t} p(x,t) = \Delta p(x,t).$$

Differential entropy

$$S[p] = -\int_X dx\, p \log p.$$

$$\nabla_{W_2} S[p] = \Delta p \longrightarrow \boxed{\frac{\partial p}{\partial t} = \nabla_{W_2} S[p]} \qquad \star$$

Heat flow is a gradient flow with respect to the entropy.

FP Equation as Wasserstein–2 Gradient Flow

General FP equation

$$\partial_t p = -\nabla \cdot (b(x)p) + \nabla \cdot (D\nabla p).$$

special form:Gradient drift

$$\partial_t p = \nabla \cdot (p\nabla V) + \Delta p, \qquad q(x) = \frac{1}{Z}e^{-V(x)}.$$

$$S(p\|q) = \int_X dx\, p(x) \log \frac{p(x)}{q(x)},$$

$$\partial_t p = -\nabla_{W_2} S(p\|q).$$

$$\frac{d}{dt}S(p_t\|q) = -\int_X p_t \left|\nabla \log \frac{p_t}{q}\right|^2 dx \le 0, \qquad \sigma(t) = \int_X p_t \left|\nabla \log \frac{p_t}{q}\right|^2 dx.$$

FP Equation as Wasserstein–2 Gradient Flow has relative entropy as a monotone.

Comment: this form requires a gradient drift $b = -\nabla V$

From FP Gradient Flow to ERG Gradient Flow

Known facts

- The ERG flow is a **functional Fokker–Planck equation**.
- The Fokker–Planck equation can be written as a **Wasserstein–2 gradient flow**.

Question

Can the ERG flow be written as some form of

$\mathcal{W}_2$-gradient flow?

Functional generalization of optimal transport

Initial and final distributions: $P_1[\phi]$, $P_2[\phi]$

Transport kernel: $\Pi(\phi_1, \phi_2)$

$$\int [d\phi_2]\Pi = P_1[\phi_1], \qquad \int [d\phi_1]\Pi = P_2[\phi_2].$$

Cost: $\mathcal{C} : \mathcal{F} \times \mathcal{F} \to \mathbb{R}$

$$K[\Pi] = \int [d\phi_1][d\phi_2]\ \Pi[\phi_1, \phi_2]\ \mathcal{C}[\phi_1, \phi_2].$$

Wasserstein–2 distance

$$\mathcal{W}_2(P_1, P_2) := \left(\inf_{\Pi \in \Gamma(P_1, P_2)} \frac{1}{2} \int [d\phi_1][d\phi_2]\ \Pi[\phi_1, \phi_2] \int d^d x\, d^d y\ \dot{C}_\Lambda(x, y) \Delta\phi_x \Delta\phi_y \right)^{1/2},$$

where $\Delta\phi_x := \phi_1(x) - \phi_2(x)$.

RG flow equation

$$\text{Distribution:} \quad P_\Lambda[\phi] = \frac{1}{Z_{P,\Lambda}} e^{-S_\Lambda[\phi]},$$

$$\text{Background:} \quad Q_\Lambda[\phi] = \frac{1}{Z_{Q,\Lambda}} e^{-2\widehat{S}_\Lambda[\phi]}.$$

$$S(P\|Q) = \int [d\phi] P[\phi] \log \left(P[\phi]/Q[\phi] \right).$$

$$\boldsymbol{-\Lambda \frac{d}{d\Lambda} P_\Lambda[\phi] = -\nabla_{W_2} S(P_\Lambda[\phi] \| Q_\Lambda[\phi])}$$

RG flow is a gradient flow with respect to the relative entropy.

Regularized relative entropy as a monotone

RG equation suggests that something like the relative entropy is monotonic under RG flow.

Define

$$M_\Lambda(P_\Lambda) := S(P_\Lambda \| Q_\Lambda) - \log(Z_{Q,\Lambda}).$$

$$\boxed{-\Lambda \frac{d}{d\Lambda} M_\Lambda(P_\Lambda) \geq 0} \longrightarrow \text{RG monotone}$$

Imply the irreversibility of the RG flow(not rigirously)

Dictionary: RG and Optimal Transport

RG concept	OT / information concept
Coarse graining	Flow of probability measures
Wegner–Morris equation	Functional continuity / FP equation
ERG kernel $\dot{C}_\Lambda$	Mobility kernel under Wasserstein metric
Seed action $\widehat{S}_\Lambda$	Prior Q_Λ
Irreversibility	regularlized Relative-entropy monotonicity

Discussion

$$-\boldsymbol{\Lambda}\frac{\boldsymbol{d}}{\boldsymbol{d\Lambda}}\boldsymbol{P_{\Lambda}[\phi]} = -\boldsymbol{\nabla_{W_2} D_{\mathrm{KL}}(P_{\Lambda}[\phi] \| Q_{\Lambda}[\phi])}$$

This solves the problem of what the RG dynamics is.

1. Connection to generative models

Inverse-RG diffusion: Cotler & Rezchikov 2023; Masuki & Ashida 2025.

$$\text{forward RG :} \quad P_{\mathrm{UV}} \to P_{\mathrm{IR}}$$

$$\text{reverse RG :} \quad P_{\mathrm{IR}} \to P_{\mathrm{UV}}$$

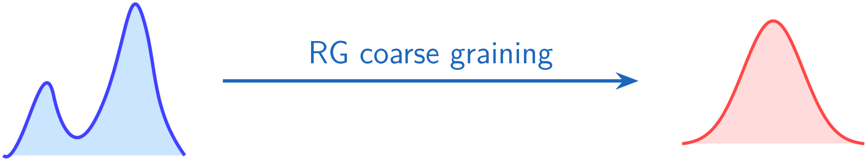
2. Bayesian framework and information geometry

$$g_{ij}(\theta) = \mathbb{E}_{\theta}[\partial_i \log p \; \partial_j \log p] = \langle O_i O_j \rangle_c$$

$$D_{\mathrm{KL}}(p_{\theta} \| p_{\theta+d\theta}) = \frac{1}{2} g_{ij} d\theta^i d\theta^j.$$

Fisher: parameter geometry;
Wasserstein: distribution geometry.

Summary



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$$-\Lambda \frac{d}{d\Lambda} P_\Lambda[\phi] = -\nabla_{W_2} D_{\mathrm{KL}}(P_\Lambda[\phi] \| Q_\Lambda[\phi])$$

RG is equivalent to the optimal transport gradient flow,give the connect between RG and information theory